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Ni_3S_2/NiFe LDH heterostructure catalysts with a built-in electric field for efficient water electrolysis

Jianfeng Wan [a], Shizheng Xie [a], Yihong Sun [a], Wenyan Bi [b,*], Qiang Zhou [b], Yikai Hou [b], Menglin Yu [a], Tianen Li [a], Dongkui Zhou [a], Baozhong Liu [b,**]

[a] School of Mechanical and Power Engineering, Henan Polytechnic University, Jiaozuo, Henan, 454000, China
[b] College of Chemistry and Chemical Engineering, Henan Polytechnic University, Jiaozuo, Henan, 454000, China

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ABSTRACT

Water cracking is a sustainable method for generating hydrogen. However, the high overpotentials during the cathodic hydrogen evolution reaction (HER) and anodic oxygen evolution reaction (OER) pose a significant obstacle to hydrogen production. This paper presents an interface engineering strategy for an efficient dual-function Ni_3S_2/NiFe LDH heterostructure electrocatalyst with a built-in electric field. By XPS investigation and DFT calculation, Ni_3S_2/NiFe LDH heterostructure forms an internal electric field at its interface, changes its electronic structure, and enhances the inherent catalytic efficiency of the catalyst, resulting in a remarkable electrochemical activity and durability in an alkaline environment. In the hydrogen extraction reaction (HER) and oxygen extraction reaction (OER), the corresponding overpotentials are 62 mV and 215 mV, respectively, at current densities of 10 mA cm^{-2}. Furthermore, this material enables it to work efficiently in alkaline electrolyzers at low cell voltages. It achieves a sustained overall water decomposition at 10 mA cm^{-2} for 120 h without degradation. This surpasses the current state-of-the-art combination of 40% Pt/C||RuO_2 in terms of performance.

1. Introduction

The global depletion of fossil resources causes a surge in energy costs, availability, and energy security. A new alternative energy is required. Hydrogen energy can satisfy the requirements due to its cleanliness, efficient energy conversion, high calorific value, and absence of carbon emissions. Producing hydrogen through water electrolysis is an effective and sustainable technology [1].

Preparing superior catalysts is pivotal in improving hydrogen production efficiency through water electrolysis [2]. The current electrocatalysts predominantly are Pt and Ru/Ir-based compositions [3,4]. However, the inadequate reserves, exorbitant expenses, and limited durability hindered the scale application. It is necessary for preparing inexpensive electrocatalysts with excellent performance [5–7], such as transition metal (hydrogen) oxides [8–10], chalcogenides [11–15], phosphides [16–18] and nitrides [19–21]. Ni and Fe hydroxides exhibit excellent OER performance [22–25], and Ni_3S_2 has outstanding HER performance. Most of these catalysts exhibit lower performance than noble metals, and many cannot sustain high performance for long in water electrolysis. Additionally, they often lack adaptability to alkaline environments.

Transition metal disulfides (TMDs) exhibit activity toward hydrogen evolution reaction [12,26–28]. So far, nickel sulfide has been widely utilized in HER, but its catalytic performance is insufficient [29]. For instance, Chen et al. reported an overpotential of 305 mV at a current density of 50 mA cm^{-2} for the Ni_3S_2@NiCo-LDH/NF catalyst [30]. Under normal circumstances, Ni_3S_2 exists alone as a catalyst with low activity, but it can be combined with other materials to form heterojunctions, and there will be excellent catalytic activity, which may be due to the change of electronic structure of metal atoms and the enhancement of electrical conductivity. The vs-Ni_3S_2/Fe_2P prepared by Ning et al. is to improve the catalytic activity of the catalyst by changing the electronic structure of the catalyst [31]. NiFe LDH exhibits better oxygen precipitation and poorer hydrogen precipitation performance [32–35]. For example, the Ni_3S_2/Cu–NiCo LDH prepared by Jia et al. presented a high HER overpotential of 156 mV at 10 mA cm^{-2} [36]. Zhao et al. prepared a Ni_3S_2@Ni_9S_8 catalyst. The inclusion of Fe and Mo effectively controls the adsorption energy of water-splitting

* Corresponding author.
** Corresponding author.
E-mail addresses: biwenyan@hpu.edu.cn (W. Bi), bzliu@hpu.edu.cn (B. Liu).

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intermediates in the electrocatalyst, leading to an increased d-band center. As a result, the electrochemical performance of Ni_3S_2@Ni_9S_8 catalyst is significantly enhanced [37]. Zhang et al. produced NiFe-LDH@NiMo-Hz@NF to alter the catalytic properties by adjusting the electronic structure of NiFe-LDH [38].

The acceleration of electron transfer can be facilitated by the presence of an internal electric field, which improves the activity of the electrocatalyst. Reconfiguring the interface between different materials or within the same material leads to an imbalanced electron distribution in that area, forming a confined electric field strength referred to as the inherent electric field (IEF) [39]. Mu et al. prepared electrocatalysts with IEF, FeO(OH)/NiFe-LDH on OER and Fe_7S_8(NiS)/NiFe-LDH on HER, where the presence of an IEF regulates the redistribution of electrons within the catalyst, while the formation of heterostructures preserves the intrinsic properties of the catalyst. The activity of the catalyst is optimized [40].

Based on the above analysis, this study prepared a novel bifunctional Ni_3S_2/NiFe LDH catalyst with a built-in electric field combined Ni_3S_2 with NiFe LDH through a two-step electrodeposition technique for efficient water decomposition. In the HER, the measured overpotentials for Ni_3S_2/NiFe LDH electrocatalysts at current densities of −10 mA cm^{-2} were 62 mV, respectively. The overpotential values for Ni_3S_2/NiFe LDH catalysts in the OER is 215 mV at current densities of 10 mA cm^{-2}, achieving a current density of 10 mA cm^{-2} with a low voltage requirement of merely 1.56 V. Density-functional theory (DFT) calculations proved that Ni_3S_2/NiFe LDH heterojunction interfaces induce an internal electric field, enhancing the inherent catalytic activity and optimizing the adsorption energy for hydrogen- and oxygen-containing intermediates (H*, OH*, and OOH*) on catalyst surfaces.

2. Experimental section

2.1. Preparation of Ni_3S_2

(1) Cleaning of Nickel Foam (NF):

Before preparation, the NF must be cleaned to remove the oxidized layer and dirt on the surface. Firstly, cut the NF into 1cm × 1 cm pieces, ultrasonic clean in a 6 mol/L hydrochloric acid solution for 20 min, anhydrous ethanol for 20 min and deionized water for 20 min. Then, vacuum dried in an oven for 12 h.

(2) Preparation of Ni_3S_2:

Adding 52.5700 g of $NiSO_4{\cdot}6H_2O$, 9.5076 g of $NiCl_3{\cdot}6H_2O$, 6.1830 g of H_3BO_3, and 29.7816 g of $Na2S_2O_3{\cdot}5H_2O$ into 200 ml of deionized water and dissolved. A 50 ml solution was measured and put into the electrolytic cell. The treated NF was introduced into an electrolytic cell with graphite rods serving as counter electrodes. The NF acted as the working electrode, while a saturated calomel electrode (SCE) was utilized as the reference electrode. The solution was stirred at a speed of 300 r/min. Deposited the catalyst under a constant voltage of −1.2V for 1 h.

2.2. Preparation of Ni_3S_2/NiFe LDH

0.5816 g of Ni $(NO_3)_2{\cdot}6H_2O$ and 0.4040 g of Fe $(NO_3)_3{\cdot}9H_2O$ dissolved in 50 ml of deionized water. The solution was transferred into an electrolytic cell featuring a three-electrode configuration, and Ni_3S_2 served as the working electrode. A voltage of −1.7 V was consistently maintained for 30 min. The Ni_3S_2/NiFe LDH was obtained by deposition and rinsed with deionized water.

2.3. Synthesis of electrodes containing 40% Pt/C and RuO_2

A mixture of 20 mg of Pt/C (or RuO_2) with a weight percentage of 40% was dispersed in a solution consisting of 180 μL pure water, 20 μL Nafion, and 200 μL anhydrous ethanol and ultrasonic for 1 h to make the solution uniform. Afterwards, the solution was evenly applied onto nickel foam by adding 40 μL and dried under vacuum conditions at 60 °C.

2.4. Materials characterization

The XRD patterns of 7 samples were analyzed using the Smart Lab X-ray diffractometer. Cu Kα radiation from a Rigaku Corporation source in Japan with a wavelength of 1.5418 A was employed. The test range for the Angle 2θ was set between 10° and 80°. Scanning electron microscopy images were captured using the Merlin Compact scanning electron microscope manufactured by Carl Zeiss NTS GmbH in Germany. EDS mapping, TEM image, and crystal fringes of Ni_3S_2/NiFe LDH were captured by the JEOL JEM 2100F high-resolution energy spectrometer in Japan. The Thermo ESCALAB 250XI X-ray photoelectron Spectrometer (XPS) and the Al k-α (1488.6 eV) X-ray source from Thermo Scientific K-Alpha were utilized to obtain the valence states of the elements.

2.5. Density functional theory (DFT) calculations

To indicate a transfer of charge from NiFe LDH to Ni_3S_2, the Vienna ab initio simulation package (VASP) was utilized to perform computational calculations [41], and the outcomes underwent further analysis utilizing the Vaspkit software [42]. The program package incorporates the GGA-PBE (Perdew-Burke-Ernzerh) method, which employs a generalized gradient exchange approximation to describe electronic interactions during structure optimization and self-consistent calculations [43]. The utilization of spin polarization is activated, while the truncation energy for kinetic energy is uniformly established at 450 eV across all plane-wave basis sets, considering exchange-correlation generalization. The Brillouin zone was determined using Monkhorst-Pack grids with dimensions of 5 × 3 × 1 for NiFe LDH (001), Ni_3S_2 (111), and Ni_3S_2/NiFe LDH. The optimization was conducted using the CG technique with Van der Waals adjustments, incorporating the DFT-D3 method approach. The energy and force parameters were fine-tuned to 1×10^{-6} and 0.02 eV A^{-1}, respectively, until achieving convergence.

2.6. Electrochemical measurements

A three-electrode configuration was employed with an electrochemical workstation (CHI760E) to determine the catalyst parameters. The working electrode is Nickel Foam with catalysts, the auxiliary electrode is a carbon rod, and the reference electrode is a saturated calomel electrode (SCE). The electrolyte solution is KOH with a pH value of 14. Nitrogen was injected into the KOH solution before the test. The working electrode area immersed in the KOH solution is 1 cm^2. To make the measurement result stable, cyclic voltammetry is used to test for 5 min before measurement to make the curve stable. The LSV curve for HER is measured at a scan rate of 2 mV s^{-1} and a voltage range of −1.7 to −1 V (vs. SCE) using a 95% IR compensation curve. The difference between the LSV curve of OER and HER gives a voltage range of −0.1 to 0.7 V (vs. SCE). Transform the LSV curve into the Tafel slope graph. Solution resistance (R_s) and charge transfer resistance (R_{ct}) are measured using the AC impedance method, with frequencies ranging from 0.01 Hz to 105 Hz, voltage applied by HER is −1.26V, voltage applied by OER is 0.46 V, and amplitude is 0.005 V. Non-Faraday regions are assessed within the voltage range of −0.4 to 0.5 V (versus SCE), employing sweep rates ranging from 10 to 100 mV s^{-1}, including intervals of 20, 40, 60, 80, and even up to a maximum of 100 mV s^{-1}. According to the CV curve, the double layer's capacitance value (C_{dl}) is calculated, and the specific surface area of electrochemical activity (ECSA) is calculated. The activity persistence was measured by step current density chronopotentiometry, and the stability of the total

solution was measured by chronopotentiometry. The equation for converting the potential of a saturated calomel electrode (SCE) to the potential of a reversible hydrogen electrode (RHE) can be represented as follows:

$$E_{RHE} = E_{SCE} + 0.059 \times pH + 0.242 \quad (1)$$

3. Results and discussion

3.1. Structural and morphological of Ni_3S_2/NiFe LDH heterostructures

The method for preparing Ni_3S_2/NiFe LDH is illustrated in Fig. 1a. Ni_3S_2/NiFe LDH was synthesized through a two-step electrodeposition process. NiFe LDH was uniformly mixed into Ni_3S_2 to form a heterostructure. Initially, Ni_3S_2 was prepared in a three-electrode system, followed by forming a heterostructure by depositing NiFe LDH based on the first step. The formation of NiFe LDH commences with the generation of hydroxyl ions through nitrate reduction. Subsequently, a negative potential is applied to the electrode, and OH^- ions can react with Ni^{2+} and Fe^{2+} ions, forming monomeric nickel hydroxide and iron hydroxide. Some divalent cobalt is oxidized to trivalent cobalt by reacting with oxygen. Additionally, due to the low solubility of Ni_3S_2, S^{2-} ions released by hydrolysis further interact with the outer surface of NiCo-LDH nanosheets, ultimately leading to the formation of Ni_3S_2/NiFe LDH heterostructures. This process causes a change in the color of the NF substrate (Fig. S1).

The X-ray diffraction (XRD) analysis reveals the structural characteristics of Ni_3S_2, Ni_3S_2/NiFe LDH, and NiFe LDH (Fig. 1b). By electrodeposition, Ni_3S_2/NiFe LDH heterostructures were prepared, showing the superposition of Ni_3S_2 and NiFe LDH phases. The X-ray diffraction patterns of Ni_3S_2/NiFe LDH and Ni_3S_2(76–1870) are similar, suggesting that the incorporation of NiFe LDH has a minimal impact on the physical phase composition of Ni_3S_2/NiFe LDH. Additionally, the absence of diffraction peaks associated with NiFe LDH was observed in Ni_3S_2/NiFe LDH.

This is because the diffraction peak positions between NiFe LDH and Ni_3S_2/NiFe LDH are close, or the content of NiFe LDH within the Ni_3S_2/NiFe LDH structure is low. No additional impurity phases were detected in the XRD pattern, suggesting that Ni_3S_2/NiFe LDH synthesis was successful. The XRD spectrum (Fig. S2) was acquired by modifying the Ni:Fe ratio during the second electrodeposition process. The spectra were most consistent when Ni:Fe is 2:1. Diffractograms were obtained for Ni_3S_2/NiFe LDH samples with varying ratios of Ni/Fe, and no noticeable variations were detected (Fig. S2).

The analysis using scanning electron microscopy (SEM) indicated the existence of a layer consisting of nanoblocks that demonstrated development on the nickel foam's surface. The SEM of NiFe LDH shows that a layer of nanoblocks grows on the surface of nickel foam without a nanosphere structure. On the surface of Ni_3S_2/NiFe LDH, it can be seen that the nanorods and nanoblocks are uniformly mixed to form a heterogeneous structure. SEM images of the second electrodeposition step with different ratios of Ni and Fe can be seen in Fig. S3. The higher the ratio of Fe, the more deposited ones formed slab-like nanoblocks. As shown in Fig. S3, Ni_3S_2 and NiFe LDH are uniformly mixed; when the content of Ni is increased, NiOH will encapsulate Ni_3S_2 and cannot play the role of Ni_3S_2, which reduces the hydrogen precipitation performance. The elemental mapping analysis of Ni_3S_2/NiFe LDH indicates a uniform distribution of Fe, Ni, O, and S across the nickel foam. It indicates that NiFe LDH nanoarrays are uniformly distributed on Ni_3S_2 nanoboxes (Fig. 2g). In Fig. S4, the mass content of four elements can be seen, of which Fe is the most important element, which may play the greatest role in the catalytic reaction. Transmission electron micrographs (Fig. 2h–i) provide evidence for the successful synthesis of a heterostructure consisting of Ni_3S_2/NiFe LDH. The lattice parameter of 0.19 nm depicted in Fig. 2i corresponds to the crystallographic plane (015) of NiFe LDH. Lattice parameters measuring 0.23 nm and 0.28 nm correspond to the mirror planes (021) and (110) of Ni_3S_2. To observe the effect of stirring solution on Ni_3S_2/NiFe LDH, increasing the rotor in the second step of electrodeposition leads to a sharp decrease in the quality of NiFe LDH deposited. As shown in Fig. S5a, fewer bright lumps of material appear, i.e., NiFe LDH. Fig. S6a shows the TEM image, in which the material marked by the blue circle is Ni_3S_2 and the material marked by the red circle is NiFe LDH. Fig. S6b displays the results of Selected Area Electron Diffraction (SAED), which revealed the presence of three crystalline facets in the Ni_3S_2/NiFe LDH catalytic material, specifically identified as (012), (110), and (202). The crystal planes (012) and (110) correspond to the NiFe LDH material, while the crystal plane (202) corresponds to the Ni_3S_2 material. The results obtained from the analysis of TEM and XRD demonstrate a congruence, offering substantiation for the effective production of the catalytic material Ni_3S_2/NiFe LDH.

The X-ray photoelectron spectroscopy (XPS) technique was employed to comprehensively analyze the composition and valence states of the electrocatalyst Ni_3S_2/NiFe LDH. The spectra obtained from the comprehensive XPS measurements indicate the existence of Ni, Fe, and S elements (Fig. 3a). The Ni 2p spectrum shown in Fig. 3b was analyzed to identify two distinct peaks corresponding to the spin-orbit doublets. The detected peaks corresponding to the $2p_{3/2}$ and $2p_{1/2}$ states were attributed to the Ni^{2+} substance, with binding energies measured at approximately 855.79 and 873.52 eV, respectively. The vibrational satellite signals corresponding to the $2p_{3/2}$ and $2p_{1/2}$ energy levels were observed at approximately 860.86 and 878.61 eV, respectively, in the Ni 2p spectra [44]. The Fe 2p spectra shown in Fig. 3c exhibit two distinct peaks at binding energies of 710.83 and 723.96 eV, which can be associated with Fe^{2+} species within the Ni_3S_2/NiFe LDH material.

Additionally, the two peaks observed at 714.29 and 727.29 eV can be attributed to the presence of Fe^{3+} material at energy levels

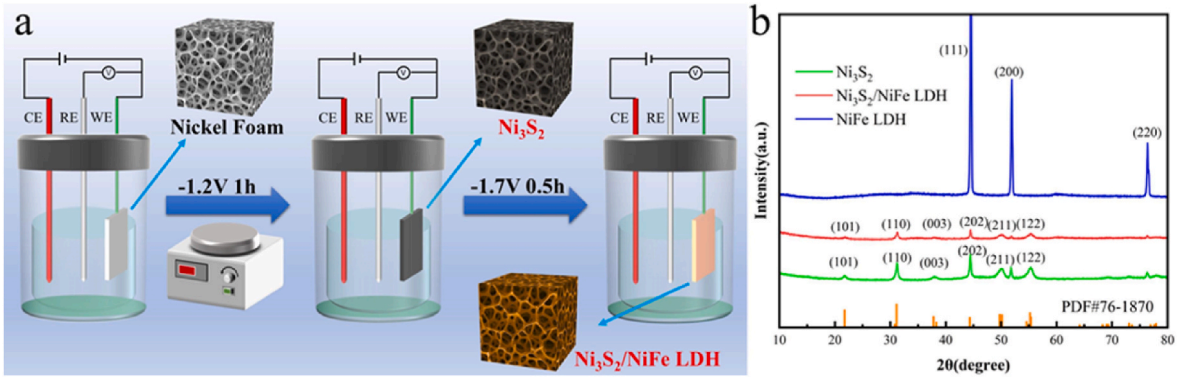


Fig. 1. (a) Depiction of the synthesis process for heterostructures comprising Ni_3S_2/NiFe LDH. (b) XRD patterns of Ni_3S_2, Ni_3S_2/NiFe LDH, and NiFe LDH.

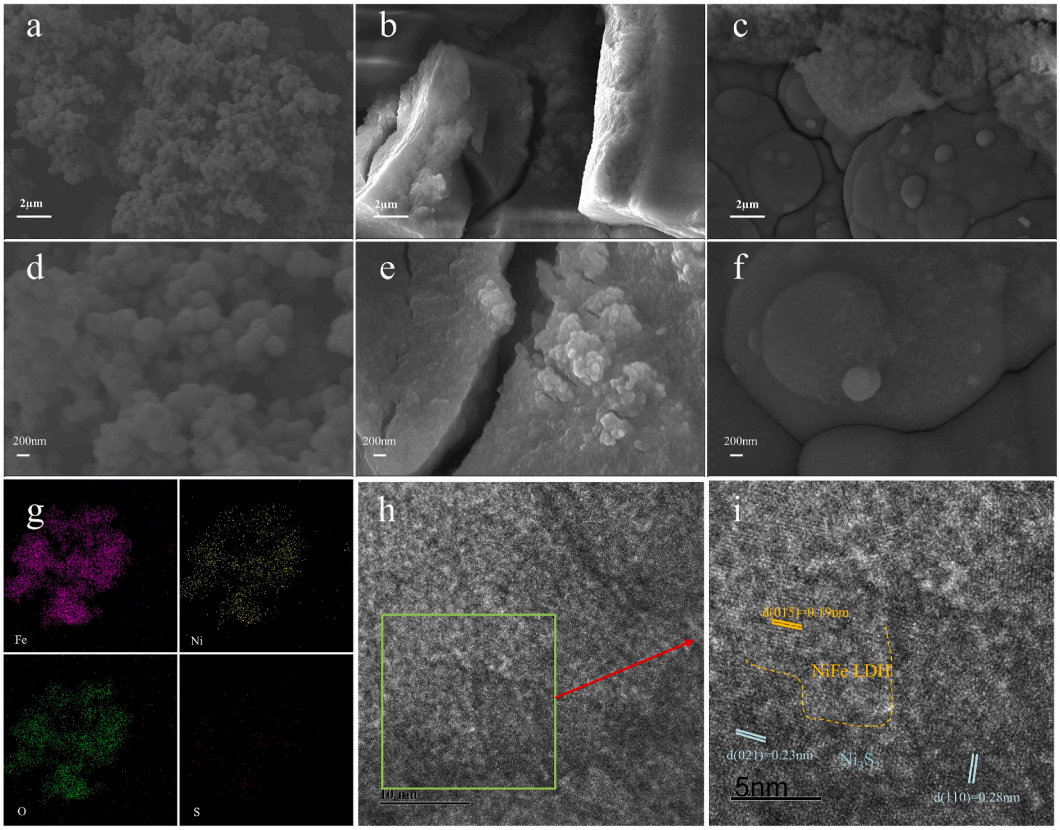


Fig. 2. SEM images of (a) Ni_3S_2, (b) NiFe LDH and (c) Ni_3S_2/NiFe LDH. (d, e, and f) are the magnified SEM images of (a), (b) and (c). (g) EDS mapping, (h) TEM image, (i) Ni_3S_2/NiFe LDH crystal fringes.

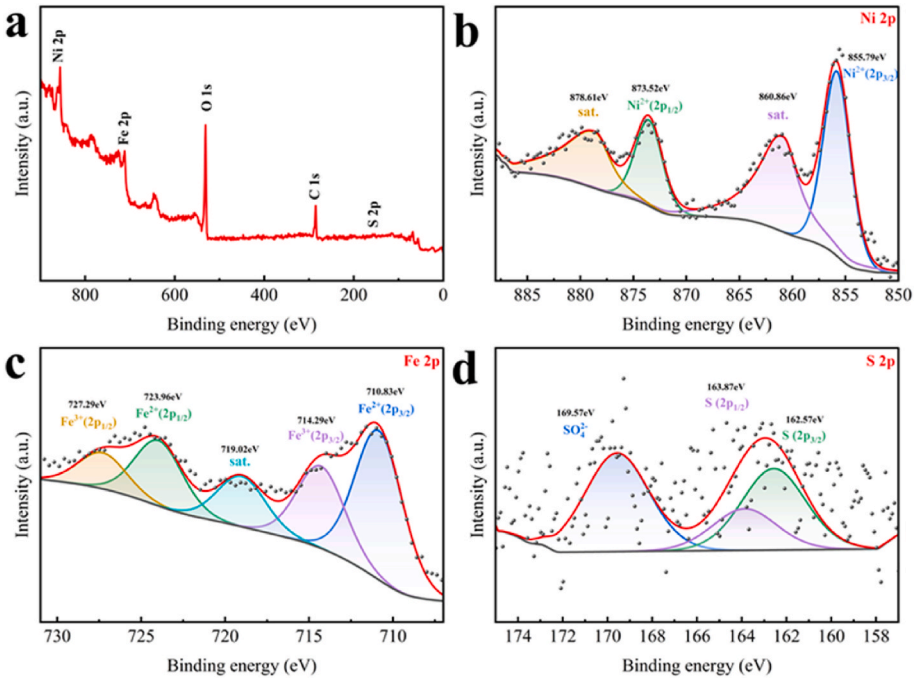


Fig. 3. Ni_3S_2/NiFe LDH-catalyzed electrodes with (a) total XPS spectra, (b) XPS spectra of Ni 2p, (c) Fe 2p, (d) S 2p.

corresponding to $2p_{3/2}$ and $2p_{1/2}$. The presence of vibrational satellite peaks is responsible for the binding energy observed at approximately 719.02 eV in the Fe 2p spectrum [45]. The S 2p spectra depicted in Fig. 3d reveal that the peaks at binding energies of 162.57 and 163.87 eV correspond to the S $2p_{3/2}$ and S $2p_{1/2}$ orbitals, respectively. The potential existence of a peak around 169.57 eV may be linked to the generation of SO_4^{2-} ions via an oxidative mechanism. The above results indicate that the Ni_3S_2/NiFe LDH catalytic material was successfully

synthesized [46–48].

The Fe content in NiFe LDH is comparatively lower than that in Ni_3S_2/NiFe LDH, suggesting a higher electron density of NiFe LDH than Ni_3S_2/NiFe LDH (Fig. S7). The transfer of electrons occurs from NiFe LDH to Ni_3S_2 in the heterostructure, creating an inherent electric field. This leads to modifications in the catalyst's electronic structure and influences its electrocatalytic OER activity.

3.2. Activity assessment based on HER

The electrocatalytic HER performance was examined in a 1 M KOH electrolyte. A comprehensive investigation was conducted to analyze the impact of the composition of the heterostructure catalyst on its efficiency in HER. The results showed that Ni_3S_2/NiFe LDH with a Ni/Fe ratio of 2:1 exhibited the best HER performance (Fig. S8). This phenomenon can be attributed to the fact that the formation of nanoblocks provides more active sites and enhances contact. Fig. 4a illustrates the corrected LSV curves for Ni_3S_2/NiFe LDH, Ni_3S_2, NiFe LDH, and 40% Pt/C. Ni_3S_2/NiFe LDH performance is better than Ni_3S_2 and NiFe LDH.

The IR-compensated and uncompensated LSV of Ni_3S_2/NiFe LDH are compared (Fig. S9a), and the compensated LSV can better reflect the actual performance of the catalyst. In industrial production, 30% KOH is generally used as the electrolyte. Fig. S14a shows the HER LSV diagram of Ni_3S_2/NiFe LDH at 30% KOH, from which it can be seen that at the current density of 10 mA cm^{-2}, HER overpotential is 6 mV respectively. The HER activity of Ni_3S_2/NiFe LDH is comparable to that of 40% Pt/C, as depicted in Fig. 4b. It demonstrates the overpotentials are 62 and 194 mV at current densities of 10 and 100 mA cm^{-2}, respectively. More importantly, it outperforms 40% Pt/C at 204 mV (see Fig. 4a). The NF exhibits a current density of 0 mA cm^{-2} before reaching 200 mV proving the influence of NF on the HER performance of Ni_3S_2/NiFe LDH is minimal. The heterostructure interfaces between Ni_3S_2 and NiFe LDH brought superior Ni_3S_2/NiFe LDH performance. The Tafel slope of Ni_3S_2/NiFe LDH is 77 mV dec^{-1}, which ranks second only to that of 40% Pt/C (36 mV dec^{-1}) and surpasses both Ni_3S_2 (96 mV dec^{-1}) and NiFe LDH (180 mV dec^{-1}). This implies that the kinetics for HER are efficient, as shown in Fig. 4c. The Nyquist plots of the equivalent circuit model can be seen in Fig. 4d for all four electrodes.

The Ni_3S_2/NiFe LDH electrode exhibits a lower charge transfer resistance (R_{ct} = 1.17Ω) than the Ni_3S_2(R_{ct} = 1.66Ω), NiFe LDH(R_{ct} = 32.63Ω), and 40% Pt/C(R_{ct} = 1.60Ω) electrodes. This suggests that during HER, there is an improvement in charge transfer kinetics, leading to a significant enhancement in the charge transfer rate. The stability tests were conducted on Ni_3S_2/NiFe LDH electrodes for 11 h. It indicates that the overpotential of Ni_3S_2/NiFe LDH remains unchanged at the same current (Fig. 4e). After conducting 1000 cycles of CV (Fig. S10a), the LSV profiles of Ni_3S_2/NiFe LDH electrodes displayed consistent characteristics, both before and after 120 h of cycling while operating at a current density of 10 mA cm^{-2} (Fig. S10b). When the current density is 200 mA cm^{-2}, Ni_3S_2/NiFe LDH is used to test the stability. It can be seen from Fig. S15 that the potential does not change at HER, indicating excellent stability. A marginal rise in overpotential was observed, indicating favorable durability and stability. The reason is that the electrode material prepared by electrochemical deposition is self-supporting, and its stability is vital. Our catalyst exhibits excellent activity, especially at 10 mA cm^{-2} current density. It outperforms many recently reported (Table S1 and Fig. 4f). From the HER currents to ECSA normalized, Ni_3S_2/NiFe LDH has larger currents and higher intrinsic activity (Fig. S11e). Furthermore, the electrochemical double-layer capacitance (Cdl) was utilized to assess the electrochemically active surface area (ECSA). The Ni_3S_2/NiFe LDH heterostructure exhibits the highest ECSA (27.9 mF cm^{-2}), which is 3.7 and 11.2 times of Ni_3S_2 (7.5 mF cm^{-2}) and NiFe LDH (2.5 mF cm^{-2}), respectively (Fig. S11d). These findings imply that the enhanced electrochemical surface area can be attributed to a heterogeneous junction surface between Ni_3S_2 and NiFe LDH. The specific surface area of the electrocatalyst is determined by N_2 adsorption/desorption measurements and provides specific activity (current density normalized by BET surface area) and TOF. The specific surface area of

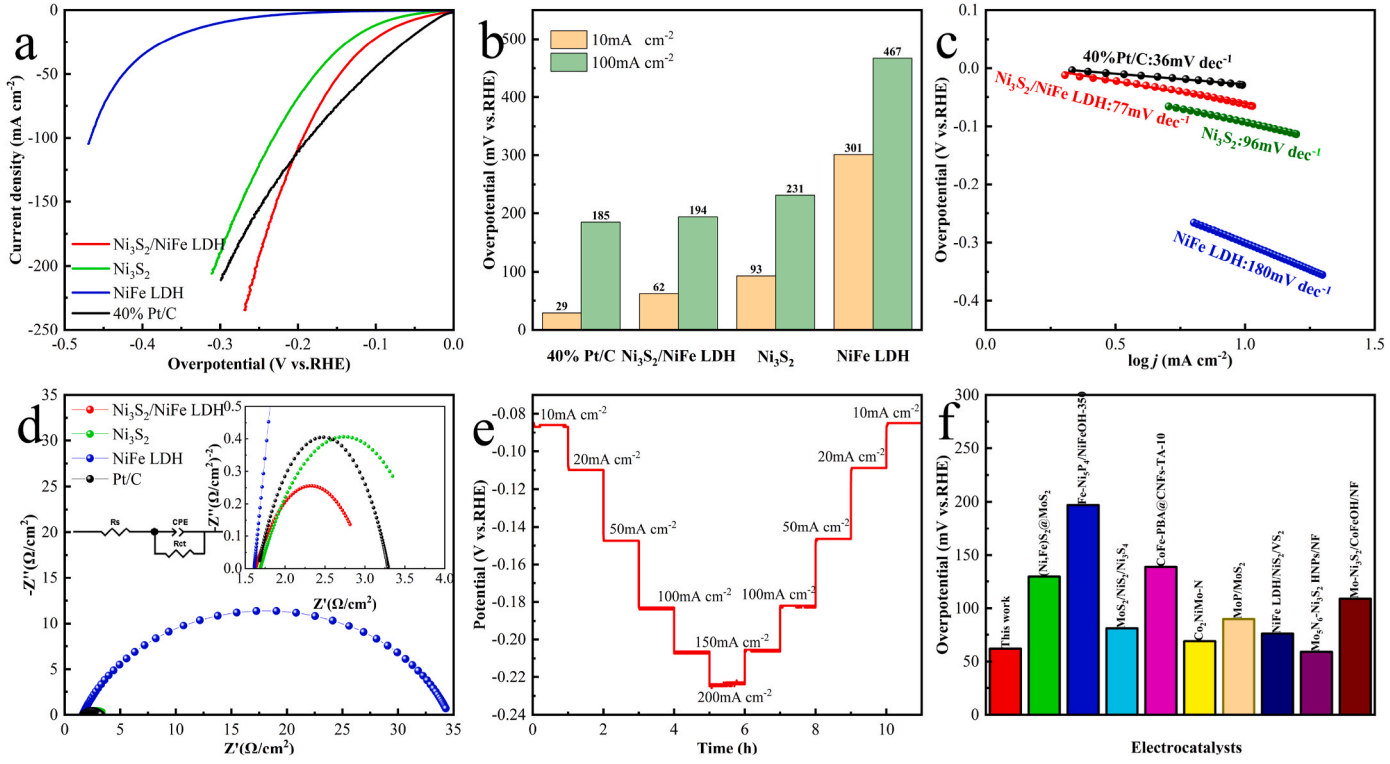


Fig. 4. Electrocatalytic HER activity of Ni_3S_2/NiFe LDH in 1 M KOH. (a) Curves of LSV, (b) Overpotential, (c) Tafel plots, (d) Nyquist plots, and (e) Chrono potentiometric curves of the Ni_3S_2/NiFe LDH collected at different current densities. (f) Assessment of the HER performance of the Ni_3S_2/NiFe LDH and diverse electrocatalysts with reported high performance at a current density of 10 mA cm^{-2}.

Ni_3S_2/NiFe LDH measured by the BET method is 1.7422 m^2/g, and the catalyst weighs 0.2941g. In Fig. S17a, the LSV curve of HER was normalized by BET surface area, and then the TOF curve was obtained by normalized current density (Fig. S17b). The TOF value of this catalyst is larger than that of many other reported catalysts, and the catalytic activity of the catalyst is excellent.

3.3. Activity assessment based on OER

The OER performance of Ni_3S_2/NiFe LDH was further analyzed. The optimal composition of the Ni_3S_2/NiFe LDH catalyst was determined, and it demonstrated superior performance in terms of OER when the ratio of Ni to Fe was 2:1 (Fig. S12). From the LSV curves (Fig. 5a), Ni_3S_2/NiFe LDH performs best, much higher than RuO_2. The NiFe LDH has a better OER performance, proving that the heterostructure surfaces of Ni_3S_2 and NiFe LDH make a contribution. From the SEM, Ni_3S_2/NiFe LDH has a larger surface area, a wider contact surface with the solution, and more active sites. Fig. S14b shows the LSV diagram of the OER of Ni_3S_2/NiFe LDH at 30% KOH, from which it can be seen that at the current density of 10 mA cm^{-2}, HER overpotential is 43 mV respectively. The overpotential exhibited by Ni_3S_2/NiFe LDH at a current density of 10 mA cm^{-2} is 215 mV, significantly lower than RuO_2 (269 mV), demonstrating its superior performance.

At a current density of 100 mA cm^{-2}, the overpotential exhibited by Ni_3S_2/NiFe LDH (252 mV) is comparatively lower than Ni_3S_2 (380 mV), NiFe LDH (255 mV), and RuO_2 (350 mV), as depicted in Fig. 5b.

Fig. S9b enables the LSVs comparison Ni_3S_2/NiFe LDH with and without IR compensation. The Tafel slope of Ni_3S_2/NiFe LDH (23 mV dec^{-1}) exhibits a lower value compared to that of Ni_3S_2 (57 mV dec^{-1}), NiFe LDH (45 mV dec^{-1}), and RuO_2(52 mV dec^{-1}). This observation implies that the OER kinetics are enhanced in the case of Ni_3S_2/NiFe LDH, as depicted in Fig. 5c. Comparing the Nyquist plots (Fig. 5d), Ni_3S_2/NiFe LDH has the lowest R_{ct} (0.385Ω) value, indicating a faster charge transfer during OER. Fig. 5e shows the overpotential plots of Ni_3S_2/NiFe LDH for 1 h running at different current densities. The overpotential of Ni_3S_2/NiFe LDH is unchanged at the same current densities, indicating that Ni_3S_2/NiFe LDH has an excellent stabilization line. The overpotentials of Ni_3S_2/NiFe LDH are unchanged after 1000 CV cycles at low current density. At high current density, the potential increase is minimal (Fig. S13). Ni_3S_2/NiFe LDH has excellent stability at a current density of 200 mA cm^{-2} (Fig. S15). The possible reason is that the oxygen generated at high current density cannot be released smoothly. Compared to alternative materials, the OER performance of Ni_3S_2/NiFe LDH demonstrates exceptional results (Fig. 5f). The Tafel slope of Ni_3S_2/NiFe LDH exhibits a lower value (Table S2). The promise of Ni_3S_2/NiFe LDH as an electrocatalyst for water splitting lies in its remarkable durability and high activity level for long periods. The LSV curve of OER was obtained by normalized BET surface area (Fig. S17c), and the TOF curve was obtained by normalized current density (Fig. S17d). It can be seen from the figure that the OER activity of the catalyst is superior.

Fig. 6 shows the OER stability of Ni_3S_2/NiFe LDH was tested at 200 mA cm^{-2} for 40 h.

From the SEM images (Fig. 6a–b), there is no significant change compared with Ni_3S_2/NiFe LDH without experiencing the stability test, exhibiting an excellent stability. In Fig. 6c, it is the total XPS spectrum tested for stability, in which Ni, Fe, and oxygen can be seen. Because the amount of S is relatively small, it is not indicated. According to the spectrum analysis of Fe (Fig. 6d), the amount of Fe^{3+} in Ni_3S_2/NiFe LDH after the stability test was significantly increased compared with Fe in Ni_3S_2/NiFe LDH without the stability test. This is probably because Fe^{2+} is oxidized to Fe^{3+} at the anode. By analyzing the spectrum of Ni, there is no significant change (Fig. 6e).

3.4. Activity assessment based on OWS

The water electrolysis performance of the catalytic electrodes was examined using a two-electrode system (Fig. 7).

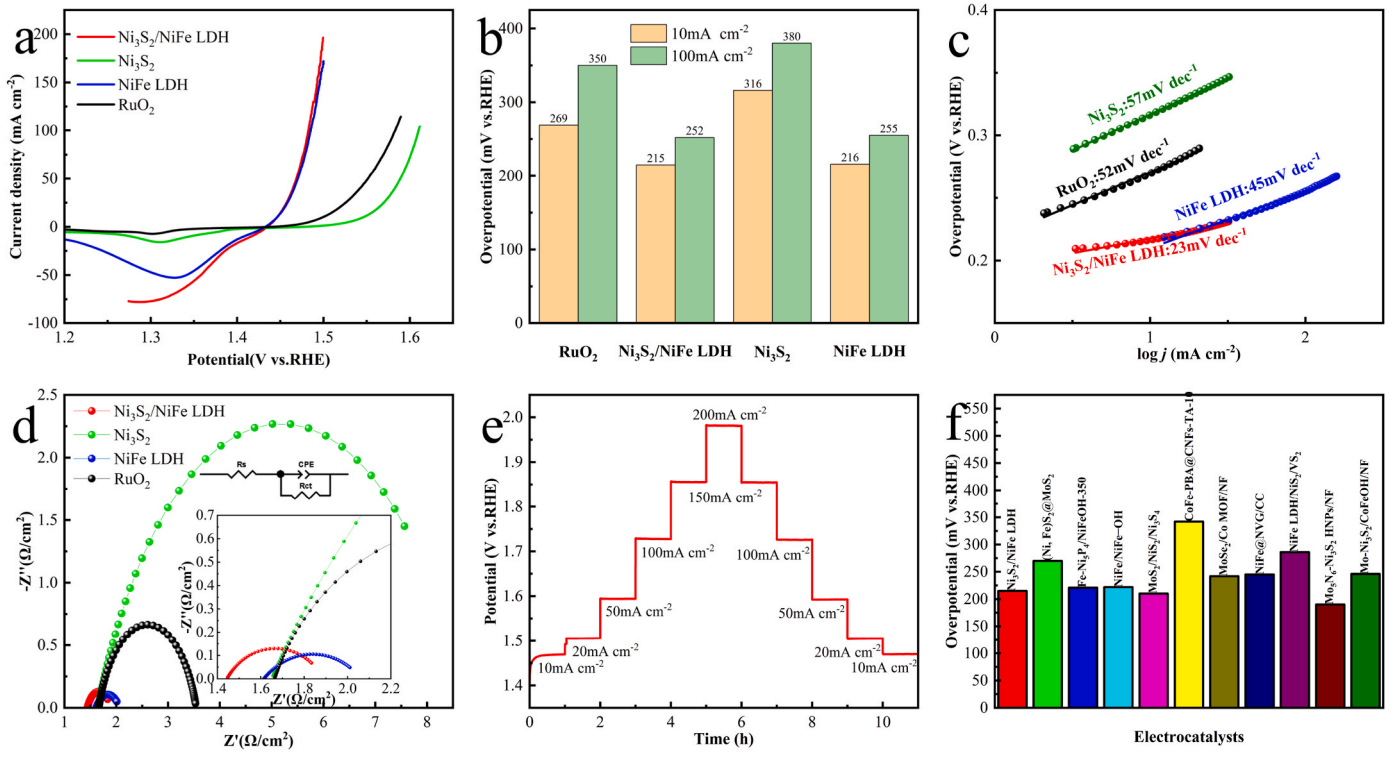


Fig. 5. Enhanced OER performance of Ni_3S_2/NiFe LDH in 1 M KOH. (a) Curves of LSV, (b) Overpotential, (c) Tafel plots, (d) Nyquist plots, and (e) Chrono potentiometric curves of the Ni_3S_2/NiFe LDH collected at different current densities. (f) Assessment of the HER performance of the Ni_3S_2/NiFe LDH and diverse electrocatalysts with reported high performance at a current density of 10 mA cm^{-2}.

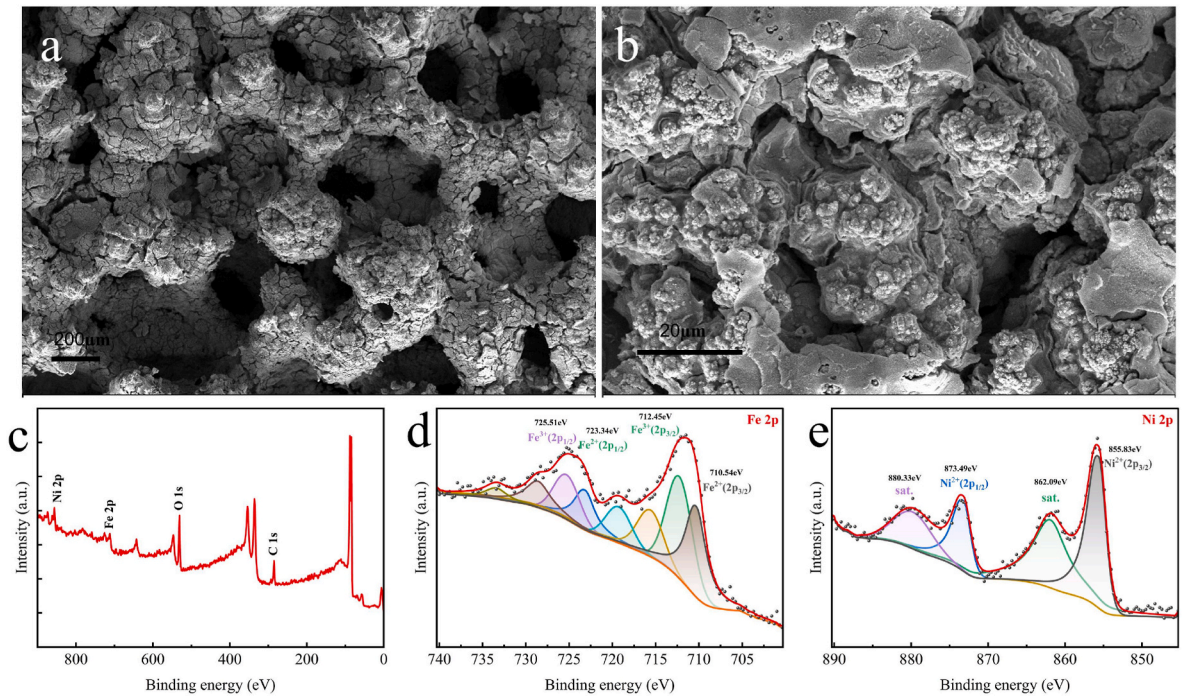


Fig. 6. The OER stability of Ni_3S_2/NiFe LDH was tested at 200 mA cm^{-2} for 40 h. (a) SEM, (b) are the magnified SEM images of (a), (c) total XPS spectra, (d) XPS spectra of Ni 2p, (e) Fe 2p.

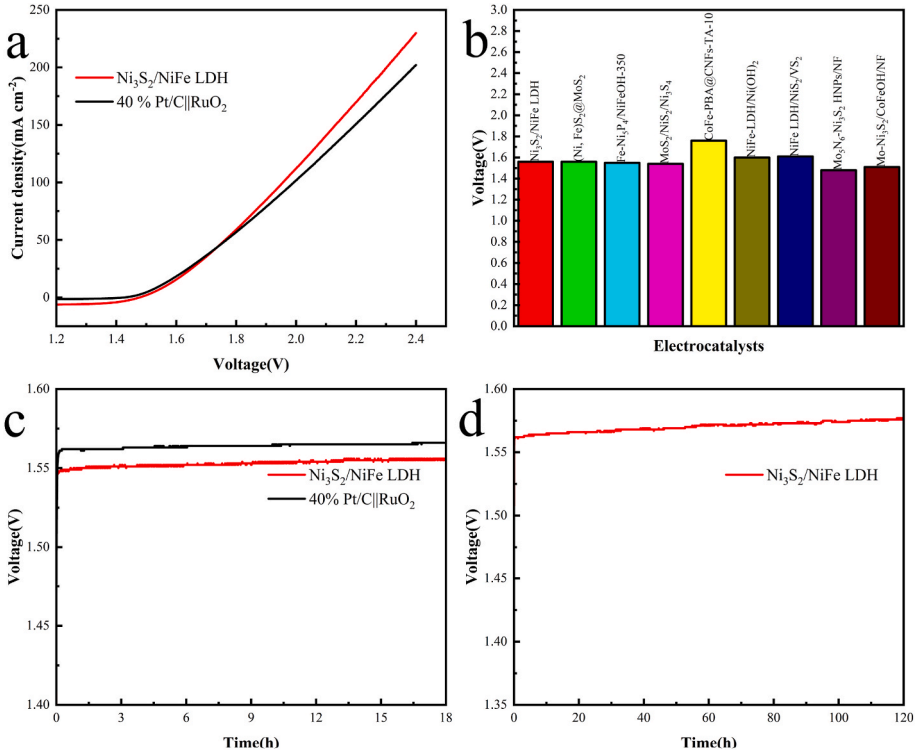


Fig. 7. Water splitting electrocatalysis was performed using the Ni_3S_2/NiFe LDH||Ni_3S_2/NiFe LDH and 40% Pt/C||RuO^2 couples in an overall manner (without IR correction). (a) LSV curves of two-electrode cells assembled by different materials, (b) Comparison of the voltage across different electrolyzes needed for a current density of 10 mA cm^{-2}, (c) Chrono-potentiometric profiles were obtained for the Ni_3S_2/NiFe LDH||Ni_3S_2/NiFe LDH and 40% Pt/C||RuO_2 systems under a current density of 10 mA cm^{-2}, (d) 120-h durability assessment of the Ni_3S_2/NiFe LDH||Ni_3S_2/NiFe LDH pair under a current density of 10 mA cm^{-2}.

Fig. 7a shows the LSV diagrams, the water electrolysis apparatus, consisting of Ni_3S_2/NiFe LDH electrodes. Due to the exceptional HER and OER properties, Ni_3S_2/NiFe LDH attains a current density of 10 mA cm^{-2} with a mere voltage input of 1.56 V. The voltages of Ni_3S_2/NiFe LDH and 40% Pt/C||RuO_2 are similar at low current density (Fig. 7a). After electrolysis for 18 h, the potential of 40% Pt/C||RuO_2 was significantly less than Ni_3S_2/NiFe LDH (Fig. 7c). To evaluate the long-term stability of Ni_3S_2/NiFe LDH, a durability test was performed in a two-

electrode electrolyze for 120 h at a current density of 10 mA cm^{-2}. In Fig. 7d, Ni_3S_2/NiFe LDH's potential rises slightly after 120 h because the material prepared in this paper is self-supporting. Ni_3S_2/NiFe LDH has a lower overall water-splitting potential than other materials (Fig. 7b). The specific potential is listed in Table S3. The water-splitting processes demonstrate remarkable efficiency and durability when using the Ni_3S_2/NiFe LDH catalytic electrode. After that, the stability of Ni_3S_2 and Ni_3S_2/NiFe LDH was tested respectively at a current density of 200 mA cm^{-2}. It can be seen from Figs. S16a–b that there was no significant change after 40 h of the stability test.

3.5. DFT calculation

We conducted DFT calculations to delve deeper into the electronic configuration of the interface between Ni_3S_2 and NiFe LDH and uncover their contributions to enhancing the catalytic performance of HER/OER. The construction of heterostructures geometrical models for Ni_3S_2, NiFe LDH, and Ni_3S_2/NiFe LDH was achieved by integrating HRTEM, XPS, and XRD findings. The presence of alkaline media enhances the HER process by facilitating catalyst adsorption and water desorption, promoting the formation of *H intermediates and subsequently enabling the generation of H_2 [49]. The assessment of the hydrogen adsorption-free energy (ΔGH*) is commonly utilized to evaluate the effectiveness of the HER. Catalysts demonstrating ΔGH* values nearing zero are considered highly encouraging candidates for promoting the HER procedure because of their capability to accelerate hydrogen liberation and proton/electron transmission [50,51].

The Gibbs free energy diagrams for hydrogen adsorption on three materials, specifically Ni_3S_2, NiFe LDH, and Ni_3S_2/NiFe LDH (as depicted in Fig. 7a), demonstrate distinct ΔGH* values (−0.67 eV, 0.40 eV, and 0.08 eV correspondingly). The hydrogen adsorption on Ni_3S_2 exhibits a robust affinity, while the hydrogen desorption on NiFe LDH exhibits a relatively weak tendency to form *H on its surface rapidly. Consequently, following Sabatier's principle, the Ni_3S_2/NiFe LDH combination possesses an optimal ΔGH* value closest to zero and offers superior free energy for efficient hydrogen adsorption.

Based on the proposed four-electron reaction mechanism by the OER [52,53], In this study, the calculations were performed to determine ΔGOH*, ΔGO* and ΔGOOH* values for the three materials (Ni_3S_2, NiFe LDH and Ni_3S_2/NiFe LDH) (Fig. 8b).

The rate-determining step (RDS) of the OER is identified by the largest free energy change. According to our findings, *OOH formation represents the highest potential barrier for Ni_3S_2 and NiFe LDH, while for Ni_3S_2/NiFe LDH, the step is *O formation. The Ni_3S_2/NiFe LDH exhibits the most favorable catalytic performance, as evidenced by its lowest ΔG of the rate-determining steps (with free energy changes of 3.11 eV, 2.55 eV, and 1.43 eV). We calculated its electronic properties as shown in Fig. 8c–f. Fig. 8c shows the schematic diagram of the internal electric field of Ni_3S_2/NiFe LDH, where electrons are transferred from NiFe LDH to Ni_3S_2 and holes are transferred from NiFe LDH to Ni_3S_2, creating an internal electric field effect that accelerates the electron transport between the interfaces and enhances the conductivity of the material. The density of states (TDOS) for NiFe LDH and Ni_3S_2/NiFe LDH heterostructures and the projected density of states (PDOS) are depicted in Fig. 8d–e. The p-orbitals of oxygen atoms form hybridized bonds with the d-orbitals of iron and nickel, leading to overlapping interactions. The primary factor responsible for the density of states (DOS) near the Fermi energy level is predominantly associated with the localized orbitals of Fe 3d and Ni 3d. The formation of the interface between Ni_3S_2 and NiFe LDH led to a noticeable alteration in the PDOS of Fe 3d. The overall upshift of the Fe 3d orbital, with its d-band center increasing from −1.14 eV to −0.78 eV and the Bader charge of the Fe ions changing from +1.5 to +1.1 optimized the free energy of the hydroxide for chemisorption, thereby improving the OER performance. To examine the influence of artificially created interfaces on the adsorption of hydrogen- and oxygen-containing intermediates, an assessment was

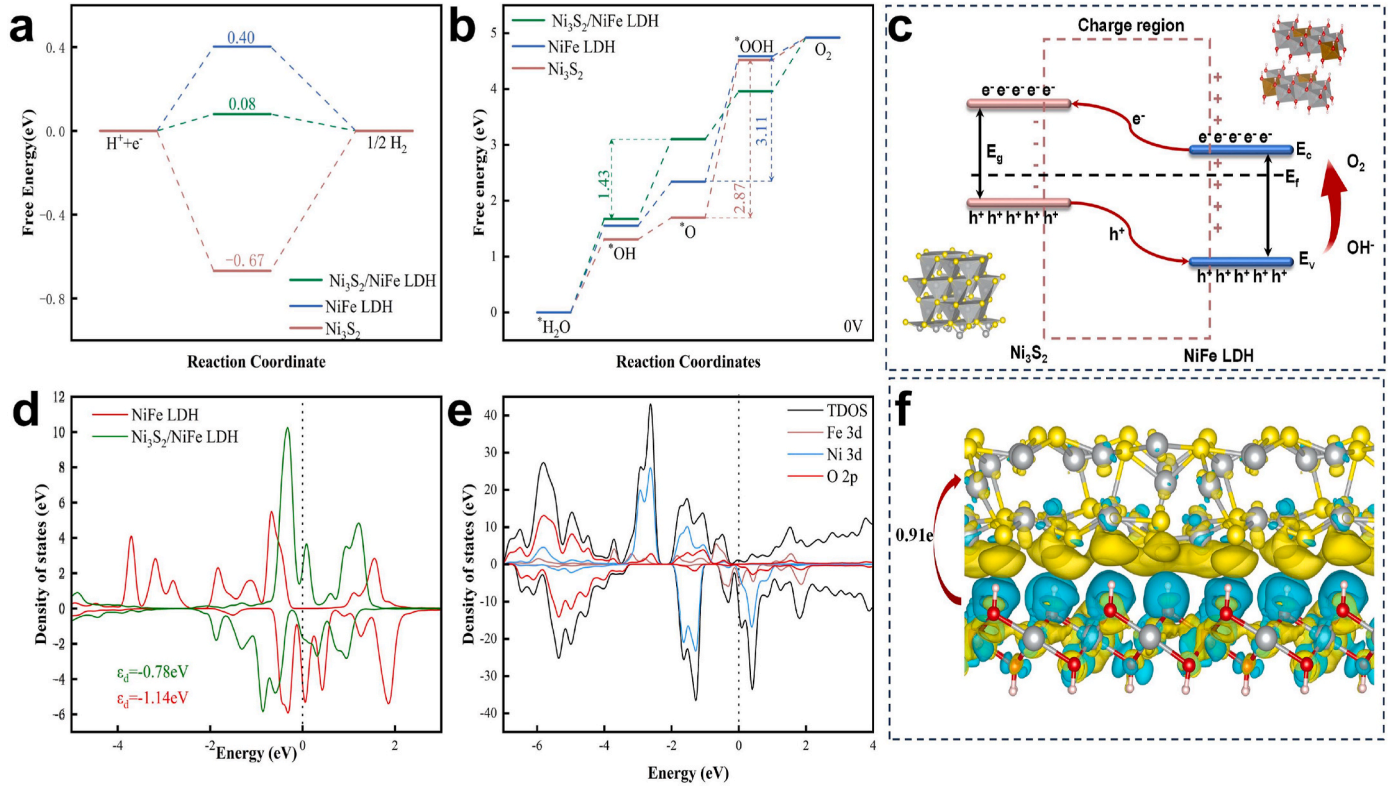


Fig. 8. (a)Hydrogen adsorption Gibbs free energy diagrams for Ni_3S_2, NiFe LDH, Ni_3S_2/NiFe LDH, (b) OER step diagrams, (c) electron transfer schematic, (d–e) projected density of states diagrams, (f) secondary charge difference density diagrams.

performed on the Bader charges at these interfaces (Fig. 8f) to determine the interlayer electronic interactions between NiFe LDH and Ni_3S_2. The results suggest a charge transfer of approximately 0.91eV occurs from NiFe LDH to Ni_3S_2. As a result, electron-rich areas are formed on Ni_3S_2, while hole-rich regions emerge on the NiFe LDH sheet. This leads to a shift in Fermi energy, which is consistent with the findings from the XPS analysis. Therefore, it can be inferred that the interaction between NiFe LDH and Ni_3S_2 significantly impacts the electrocatalyst's performance.

4. Conclusion

This paper used a two-step electrodeposition method to prepare Ni_3S_2/NiFe LDH self-supported electrocatalysts featuring clusters with bifunctional properties. The outstanding catalytic performance of the bifunctional catalysts, composed of Ni_3S_2 and NiFe LDH, could be attributed to their distinctive nanostructures, abundant catalytic sites, and the internal electric field effect. The results showed that the prepared Ni_3S_2/NiFe LDH exhibited excellent activities for HER (62 mV) and OER (215 mV) at a -10 mA cm^{-2}, respectively, and the overpotentials at low current densities were much lower than those similar catalysts. In addition, utilizing Ni_3S_2/NiFe LDH in the water electrolyze enabled a remarkable hydrolysis current of 10 mA cm^{-2} at a mere cell voltage of 1.56 V. Its performance degradation was negligible for a minimum duration of 120 h. These findings suggested that by strategically designing the electrocatalysts and identifying the Ni_3S_2/NiFe LDH catalyst, there was potential for enhancing the performance and stability of catalysts used in HER and OER. This could pave the way for its promising application in industrial hydrogen production through water electrolysis.

CRediT authorship contribution statement

Jianfeng Wan: Writing – review & editing, Methodology, Conceptualization. **Shizheng Xie:** Writing – original draft, Investigation, Formal analysis, Data curation. **Yihong Sun:** Software. **Wenyan Bi:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Qiang Zhou:** Validation. **Yikai Hou:** Validation. **Menglin Yu:** Software. **Tianen Li:** Visualization. **Dongkui Zhou:** Resources. **Baozhong Liu:** Project administration, Funding acquisition.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijhydene.2025.01.455.

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